

Swinging Sign

Stage 4 Technology · Materials and production processes · Five-week local unit

SOURCE DURATION 5 weeks		SOURCE TIMETABLE 5 x 60 min / fortnight	PROGRAM MAP 15 flexible learning blocks	RELEASE STATE Release authorised; verification pending
Course	Stage 4 Technology		Focus	Materials and production processes
Cohort	Stage 4 / Years 7-8		Delivery year	Teacher to confirm
Primary project	Swinging Sign		Additional pathways	LED tea-light holder; scrolled pot-plant holder
Syllabus	Technology 7-8 Syllabus (2023); implemented from 2026		Assessment	Orientation only; formal details Teacher to confirm
Timetable boundary: The source wording is retained exactly: 5 weeks (5 x 60-minute lessons per fortnight). W1-A to W5-C are flexible learning blocks, not a claimed whole-number period count. The teacher must confirm how they fit the local timetable.				

Program rationale

This source-first unit develops design, communication, planning, materials and production-process reasoning through the Swinging Sign and teacher-directed additional pathways. Every named section connects theory, a purposeful visual, a complete media or no-stream pathway, an applied activity, 10 feedback-rich multiple-choice questions and one scaffolded higher-order response. Practical work begins only after the approved local plan, demonstration, controls and authorisation are in place.

Stage 4 outcomes

Outcome	Exact outcome wording
TE4-SDP-01	explains relationships between sustainability, design and production
TE4-PDP-01	describes the practices and processes of designers and producers
TE4-MSD-01	explains how materials, systems and components contribute to solutions
TE4-DES-01	communicates and evaluates design ideas and solutions
TE4-PPM-01	applies processes in the planning, management and production of projects
TE4-SAF-01	selects and safely uses tools, materials, technologies and processes

Source and authority notes: **Technology 7-8 Syllabus (2023)** controls the course framework; **Stage 4 outcomes** and **Materials and production processes content** control syllabus wording. **Teacher Drive unit sources** guide the five-week sequence. This school-developed program is NESA-aligned; it is not NESA-approved or endorsed.

WEEKS 1-2 • MODULES 1-2

Teaching and learning sequence

Six flexible blocks establish the brief, readiness and design communication. Combine or distribute them only after the local timetable is confirmed.

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
W1-A <i>flexible learning block</i>	Site/theory	Read the product and user	Module 1: Read the product and user — Introduce purpose, user, context, form and function. Model the difference between visible observation and unsupported inference using privacy-safe product references. Evidence detective — complete the concept-linked applied activity. Knowledge check and scaffolded response; save or export evidence as directed.	Describe products and systems created by designers, producers and manufacturers Outline factors affecting the design of products and solutions	Four-note evidence set, revised product-analysis sentence, 10 feedback-rich MCQs and the scaffolded analysis response. My folio evidence Outcomes: TE4-PDP-01, TE4-DES-01
W1-B <i>flexible learning block</i>	Project/folio	Criteria, constraints and project pathways	Module 1: Criteria, constraints and project pathways — Model a source-safe brief and sort criteria, constraints and confirmation questions. Compare the primary Swinging Sign with teacher-directed LED tea-light and pot-plant holder pathways without inventing specifications. Pathway brief studio — complete the concept-linked applied activity. Knowledge check and scaffolded response; save or export evidence as directed.	Outline factors affecting the design of products and solutions Apply design and creative thinking to assess ideas and quality solutions	Draft brief, labelled criteria and constraints, pathway trade-off, 10 MCQs and scaffolded justification. My folio evidence Outcomes: TE4-DES-01, TE4-PPM-01
W1-C <i>flexible learning block</i>	Site/theory	Workshop readiness and sustainability	Module 1: Workshop readiness and sustainability — Explain hazard, risk, control, authorisation and the pause-and-ask decision. Trace sustainability across sourcing, production, use, maintenance and end of life; local controls remain teacher supplied. Decision gate: ready, responsible, recorded — complete the concept-linked applied activity. Knowledge check and scaffolded response; save or export evidence as directed.	Compare sustainable sourcing practices Explain ethical and legal considerations for innovation, design and production processes Demonstrate safe practices when selecting and using materials, technologies and processes	Readiness decision sheet, justified sustainability choice, 10 MCQs and scaffolded response. My folio evidence Outcomes: TE4-SDP-01, TE4-SAF-01
W2-A <i>flexible learning block</i>	Site/theory	Vector and bitmap	Module 2: Vector and bitmap — Compare how vector and bitmap graphics store and scale images. Students inspect editing evidence, select suitable formats and explain why automatic tracing still needs review. Format triage — complete the concept-linked applied activity. Knowledge check and scaffolded response; save or export evidence as directed.	Use a range of sketching and drawing techniques to communicate ideas and solutions Communicate the development of design ideas and solutions, using annotations	Format triage with four justifications, tracing-risk note, 10 MCQs and scaffolded comparison. My folio evidence Outcomes: TE4-DES-01, TE4-PDP-01

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
W2-B <i>flexible learning block</i>	Project/folio	Onshape sketches and features	Module 2: Onshape sketches and features — Model how a purposeful sketch plane, dimensions and geometric constraints capture design intent, then connect the controlled sketch to a basic Onshape feature and versioned review workflow. Onshape constraint clinic — complete the concept-linked applied activity. Knowledge check and scaffolded response; save or export evidence as directed.	Use a range of sketching and drawing techniques to communicate ideas and solutions Communicate the development of design ideas and solutions, using annotations Document design processes when using materials and production technologies	Onshape constraint-clinic observations, dimension and constraint reasons, sketch-to-feature explanation, review note, 10 MCQs and scaffolded workflow response. My folio evidence Outcomes: TE4-DES-01, TE4-PPM-01
W2-C <i>flexible learning block</i>	Project/folio	Legibility, annotation and peer review	Module 2: Legibility, annotation and peer review — Test visual hierarchy in a teacher-approved viewing routine. Model feature-reason-effect annotations and notice-connect-suggest peer feedback linked to criteria, then revise one design decision. Legibility lab — complete the concept-linked applied activity. Knowledge check and scaffolded response; save or export evidence as directed.	Apply design and creative thinking to assess ideas and quality solutions Communicate the development of design ideas and solutions, using annotations Work collaboratively to test, modify and improve the quality of ideas and solutions	Reading-order record, criteria-based feedback slip, annotated revision, 10 MCQs and scaffolded evaluation. My folio evidence Outcomes: TE4-DES-01, TE4-PDP-01

Whole-course teaching controls · access and capability

Adjustments and differentiation: Use the complete non-video pathway, chunk long responses through the three scaffold prompts, pre-teach vocabulary, provide worked examples, allow speech-to-text or oral rehearsal where appropriate, and extend students through cross-pathway comparison. Adjustments support access without lowering the evidence target.

Literacy, numeracy and digital demands: Literacy centres on precise observation, feature-reason-effect annotations, criteria-based judgements and extended explanation. Numeracy includes scale awareness, measurement language, comparison, simple area/yield reasoning and evidence limits. Digital work includes vector/bitmap concepts, Onshape sketches and drawings, constraints, version control, file review and browser-local evidence backup.

Sustainability and resource decisions: Track material quantity, layout efficiency, connected offcuts, durability, maintenance and end-of-life thinking. Sustainability claims require evidence; recyclability alone is not enough. The unit does not claim the Aboriginal and Torres Strait Islander Peoples content point without an approved local Cultural source and protocol.

WEEKS 3-4 • MODULES 3-4

Teaching and learning sequence

Six flexible blocks move from material and approval reasoning into production control and evidence. Practical work remains conditional on current local authority.

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
W3-A <i>flexible learning block</i>	Site/theory	Material properties and selection	Module 3: Material properties and selection — Connect material properties to confirmed purpose, environment and appearance. Compare only teacher-approved options and make a provisional selection that identifies evidence, a trade-off and remaining unknowns. Build a property decision matrix — complete the concept-linked applied activity. Knowledge check and scaffolded response; save or export evidence as directed.	Identify the characteristics and properties of materials Describe how the properties of materials and production techniques contribute to the quality of solutions Justify the selection and use of a range of tools, materials, techniques and technologies	Property decision matrix, provisional recommendation, 10 MCQs and scaffolded material justification. My folio evidence Outcomes: TE4-MSC-01, TE4-DES-01
W3-B <i>flexible learning block</i>	Project/folio	Measurement, layout and resource efficiency	Module 3: Measurement, layout and resource efficiency — Use paper layouts and simple area or yield reasoning to compare resource efficiency. Separate design intent from teacher-confirmed dimensions, spacing, margins, kerf and stock facts. Paper nesting challenge — complete the concept-linked applied activity. Knowledge check and scaffolded response; save or export evidence as directed.	Outline factors affecting the design of products and solutions Compare sustainable sourcing practices Document design processes when using materials and production technologies	Two annotated layouts, comparison calculation or estimate, 10 MCQs and scaffolded efficiency response. My folio evidence Outcomes: TE4-SDP-01, TE4-PPM-01
W3-C <i>flexible learning block</i>	Project/folio	Onshape drawings, mounting and approval	Module 3: Onshape drawings, mounting and approval — Assemble a reviewable approval pack: current Onshape drawing, version record, criteria, mounting concept and unresolved questions. Make the teacher approval gate explicit before any export or practical production. Audit a fictional approval pack — complete the concept-linked applied activity. Knowledge check and scaffolded response; save or export evidence as directed.	Outline factors affecting the design of products and solutions Communicate the development of design ideas and solutions, using annotations Document design processes when using materials and production technologies	Onshape drawing-and-approval audit, three review questions, version record, 10 MCQs and scaffolded proposal response. My folio evidence Outcomes: TE4-DES-01, TE4-PPM-01

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
W4-A <i>flexible learning block</i>	Site/theory	Production sequence and authorisation	Module 4: Production sequence and authorisation — Build a dependency-based production sequence with roles and hold points. Distinguish student planning decisions from teacher or authorised-person decisions; do not publish machine procedures. Build a production-control board — complete the concept-linked applied activity. Knowledge check and scaffolded response; save or export evidence as directed.	Outline production techniques and materials used by designers, producers and manufacturers Demonstrate safe practices when selecting and using materials, technologies and processes Document design processes when using materials and production technologies	Production-control board, hold-point explanation, 10 MCQs and scaffolded sequence response. My folio evidence Outcomes: TE4-PDP-01, TE4-PPM-01, TE4-SAF-01
W4-B <i>flexible learning block</i>	Practical	Edges, finishing and quality	Module 4: Edges, finishing and quality — Before any teacher-approved making, model neutral inspection language for edge and surface condition. During authorised practical work, students follow the demonstrated local method and record quality decisions; otherwise use the complete image-based pathway. Turn opinions into quality evidence — complete the concept-linked applied activity. Knowledge check and scaffolded response; save or export evidence as directed.	Describe how the properties of materials and production techniques contribute to the quality of solutions Demonstrate safe practices when selecting and using materials, technologies and processes Use factors affecting design to evaluate the quality of ideas and solutions	Inspection notes, teacher checkpoint, permitted process evidence, 10 MCQs and scaffolded quality judgement. My folio evidence Outcomes: TE4-PDP-01, TE4-SAF-01
W4-C <i>flexible learning block</i>	Project/folio	Nesting, offcuts and process evidence	Module 4: Nesting, offcuts and process evidence — Compare planned and actual layout evidence, classify useful offcuts through the local system, and select privacy-safe process evidence that explains decisions rather than decorating the folio. Edit a process-evidence story — complete the concept-linked applied activity. Knowledge check and scaffolded response; save or export evidence as directed.	Compare sustainable sourcing practices Document design processes when using materials and production technologies Use factors affecting design to evaluate the quality of ideas and solutions	Five-item evidence story, captions, future layout test, 10 MCQs and scaffolded reflection. My folio evidence Outcomes: TE4-SDP-01, TE4-PPM-01

WEEK 5 • MODULE 5

Teaching and learning sequence

Three flexible blocks connect finish systems, controlled use testing and evaluation. Use the complete model or image pathway whenever practical approval is absent.

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
W5-A <i>flexible learning block</i>	Practical	Surface preparation and coatings	Module 5: Surface preparation and coatings — Explain the functions and compatibility of substrate, preparation, primer, topcoat and intended environment. Any preparation or coating occurs only with teacher-approved products, SDS, controls and demonstration; otherwise use the fictional finish-system audit. Audit a fictional finish system — complete the concept-linked applied activity. Knowledge check and scaffolded response; save or export evidence as directed.	Identify the characteristics and properties of materials Describe how the properties of materials and production techniques contribute to the quality of solutions Justify the selection and use of a range of tools, materials, techniques and technologies	Finish-system map, inspection statement, teacher checkpoint, 10 MCQs and scaffolded justification. My folio evidence Outcomes: TE4-MSC-01, TE4-SAF-01
W5-B <i>flexible learning block</i>	Practical	Hanging, stability and use	Module 5: Hanging, stability and use — Use a teacher-provided model to trace conceptual load paths, balance and stability. Actual hanging, loading, installation or use testing requires an approved criterion and teacher control; LED tea-light language remains the default. Reason with a cardboard balance model — complete the concept-linked applied activity. Knowledge check and scaffolded response; save or export evidence as directed.	Describe products and systems created by designers, producers and manufacturers Create prototypes, models and samples to test materials and production processes Use factors affecting design to evaluate the quality of ideas and solutions	Annotated model, prediction, approved test evidence where authorised, 10 MCQs and scaffolded systems response. My folio evidence Outcomes: TE4-MSC-01, TE4-DES-01, TE4-SAF-01
W5-C <i>flexible learning block</i>	Project/folio	Evaluate, feedback and improve	Module 5: Evaluate, feedback and improve — Evaluate the product or teacher-provided case against confirmed criteria. Distinguish description, evidence and judgement; triage peer feedback and justify one feasible, testable future improvement. Feedback triage and improvement pitch — complete the concept-linked applied activity. Knowledge check and scaffolded response; save or export evidence as directed.	Work collaboratively to test, modify and improve the quality of ideas and solutions Evaluate ideas and solutions using written, visual, verbal or multimodal communication forms Use factors affecting design to evaluate the quality of ideas and solutions	Criteria-based evaluation, feedback triage, improvement pitch, 10 MCQs and scaffolded capstone response. My folio evidence Outcomes: TE4-SDP-01, TE4-DES-01, TE4-PPM-01

Whole-course teaching controls • implementation

Safety, workplace and practical authority: The website teaches readiness and decision-making, not operational procedures. Local plans, demonstrations, SDS, equipment instructions, supervision, PPE, controls and permissions remain teacher-controlled. W4-B, W5-A and W5-B are Practical only when the affected work is approved; otherwise teach the complete theory, visual, model or image-based equivalent.

Assessment and feedback boundary: Knowledge checks, long responses, activities and folio entries are formative learning evidence. No approved notification, rubric, weighting, marks, due date or submission route was supplied. The teacher links the formal assessment material before representing any task as assessed.

Project-pathway differentiation: Swinging Sign is the primary pathway. The LED tea-light holder and scrolled pot-plant holder are alternate or additional pathways only when the teacher confirms their role and plan. Use LED tea lights only unless a different policy is formally approved; loading and stability criteria for the pot-plant holder remain Teacher to confirm.

Source, privacy and release boundary: A privacy-safe hero replaces the personalised Swinging Sign photograph. Supplied holder photographs are teacher-authorised for this course release only; ownership is not asserted and third-party reuse is not granted. Steve authorised GitHub Pages release on 31 August 2026; live verification follows deployment.

CONTROL REGISTER

Teacher to confirm before use

These are bounded local decisions, not missing student theory. Confirm only the affected field and keep all other teaching moving.

ID	Decision or evidence needed	Owner / affected scope	Due before / smallest action
TTC-01	Whole-number lesson count and local rotation or timetable placement for the source wording: 5 weeks (5 x 60-minute lessons per fortnight).	Class teacher / faculty All flexible learning blocks	Timetabling and class issue Record the approved period allocation and how the 15 learning packages will be combined or distributed.
TTC-02	Whether the LED tea-light holder and scrolled pot-plant holder are alternatives, extensions or separately assessed projects.	Class teacher / faculty Project pathway selection	Students select or receive a pathway Confirm the role of each additional project and communicate it to students.
TTC-03	Approved plans and the relationship between the photographed Swinging Sign and the Drive CNC metal-sign brief.	Class teacher / faculty Project planning and practical work	Any production approval Issue the current approved plan or job sheet for each in-use pathway.
TTC-04	Approved materials, stock form, sizes, quantities, tolerances and any loading criteria.	Class teacher / faculty Modules 3-5 and every practical block	Material selection or layout becomes a production document Attach the current material and stock specification to the approval pack.
TTC-05	Tools, machines, student/teacher operating roles, demonstrations, supervision, local risk controls and evidence permissions.	Class teacher / faculty Every practical block	Each affected practical session Confirm the current local procedure and explicit authorisation checkpoint.
TTC-06	Swinging Sign mounting system, wind/load context, installation boundary and controlled test criteria.	Class teacher / faculty W3-C and W5-B	Mounting approval or any use test Issue the approved mounting and testing criteria; otherwise retain model-only learning.
TTC-07	Tea-light policy and approved holder design. The current course uses LED tea lights only unless a different policy is formally confirmed.	Class teacher / faculty LED tea-light pathway	Pathway approval and any function check Confirm the LED-only rule or provide the authorised alternative policy and plan.
TTC-08	Pot-plant holder loading, pot size, stability, support and approved testing criteria.	Class teacher / faculty Pot-plant holder pathway	Design approval or any loading test Record the approved design load and controlled test method; otherwise retain model-only learning.

ID	Decision or evidence needed	Owner / affected scope	Due before / smallest action
TTC-09	Formal assessment notification, outcomes, rubric, marks or weighting, due date, submission route and evidence requirements.	Faculty / assessment coordinator Assessment and formal submission	Any formal assessment advice is issued Link the approved notification; until then keep the website and program as assessment orientation only.
TTC-10	Onshape document sharing, class folder, file naming, units, drawing template, dimension and view conventions, export and production approval workflow.	Class teacher / faculty Modules 2-4	Onshape drawing or export submission for production review Issue the current file checklist and approval state labels.

EDITABLE IMPLEMENTATION RECORD

Delivery register

The sequence uses flexible learning blocks, not periods. Complete fields during delivery and record any combining, substitution or variance.

Planned block	Planned focus	Actual date	Initials	Completion / variance	Evidence checked	Adjustments	Follow-up
W1-A	Read the product and user						
W1-B	Criteria, constraints and project pathways						
W1-C	Workshop readiness and sustainability						
W2-A	Vector and bitmap						
W2-B	Onshape sketches and features						
W2-C	Legibility, annotation and peer review						
W3-A	Material properties and selection						
W3-B	Measurement, layout and resource efficiency						
W3-C	Onshape drawings, mounting and approval						
W4-A	Production sequence and authorisation						
W4-B	Edges, finishing and quality						
W4-C	Nesting, offcuts and process evidence						
W5-A	Surface preparation and coatings						
W5-B	Hanging, stability and use						
W5-C	Evaluate, feedback and improve						